

QUAID-I-AZAM UNIVERSITY
DEPARTMENT OF COMPUTER SCIENCE (BS 5th)

Terminal Examination Fall, 2023

Course No. ST-101 Course Title: Probability and Statistics

Max. Marks: 50

Teacher's Name: Mr. Abdullah

Iqbal Time Allowed: 3 Hours

Date: 30-JAN-2024

Note: Attempt all questions. All questions carry equal marks.

Q.1 (a) What are the values of β_1 and β_2 in a symmetrical distribution. (03)

(b) Calculate the Bowley's co-efficient of skewness from the following distribution. (07)

Ages (years)	15-19	20-24	25-29	30-34	35-39	40-44	45-49	50-54
No. of Men	29	176	208	173	82	40	15	3

Q.2 (a) What is meant by conditional probability? Briefly explain Simple and Compound Events? (02)

(b) A professor of statistics has two boxes of computer disks: box C_1 contains seven Verbatim disks and three Control Data disks and box C_2 contains two Verbatim disks and eight Control Data disks. She selects a box at random with probabilities $P(C_1) = 2/3$. And $P(C_2) = 1/3$ because of their respective locations. A disk is then selected at random and the event C occurs if it is from Control Data, Using an equally likely assumption for each disk in the selected box, compute $P(C_1 | C)$ and $P(C_2 | C)$. (03)

(c) Three distinct integers are chosen at random from the first 20 positive integers. Compute probability that: (04)

(i) their sum is even; (ii) their product is even

Q.3 (a) Define correlation and product moment co-efficient of correlation. (02)

(b) Given the following set of values:

Advertising Cost	20	11	15	10	17	19
Weekly Sales	5	15	14	17	8	9

Determine the equation of the least square regression line of Y on X, to predict the Weekly Sales? (04)

(c) Calculate the correlation co-efficient by first multiplying each X and Y by 10 and then subtracting 70 from each X and 60 from each Y. (04)

X	8.2	9.6	7.0	9.4	10.9	7.1	9.0	6.6	8.4	10.5
Y	8.7	9.6	6.9	8.5	11.3	7.6	9.2	6.3	8.4	12.4

Q.4 (a) The probability of surviving a certain transplant operation is 0.55. If a patient survives the operation, the probability that his or her body will reject the transplant within a month is 0.20. What is the probability of surviving both of these critical stages? (03)

(b) A manufacturer of cotton pins knows that 5% of his product is defective. If he sells cotton pins in boxes of 100, and guarantees that no more than 4 pins will be defective, what is the probability that a box will fail to meet the guaranteed equality? (04)

(c) A committee of size 5 is to be selected at random from 3 women and 5 men. Find the probability that the committee consists of (i) All men (ii) All women. (03)

Q.5 (a) Differentiate between joint, marginal and conditional probability. (03)

(b) The following table shows the distribution of random variable X (03)

x	1	2	3	4
f(x)	1/10	3/10	A	1/10

Find (i) the value of A. (ii) mean and variance of X. (04)

(c) Let X and Y have the joint probability functions given by: $f(x, y) = \frac{xy}{66}$ for $x = 2, 4, 5$ and $y = 1, 2, 3$ (04)

Find marginal probability functions of X and Y, and find out if X and Y are independent (03)

Q12-

a) $\beta_1 = 0$, β_2 varies in symmetrical

b) Bowley's coeff. of skewness = $\frac{Q_3 + Q_3 - 2 \text{ Median}}{Q_3 - Q_1}$

$$Q_1 = L + \frac{h}{f} \left(\frac{n}{4} - c \right), \quad Q_3 = L + \frac{h}{f} \left(\frac{3n}{4} - c \right)$$

$$\text{Median} = L + \frac{h}{f} \left(\frac{n}{2} - c \right)$$

→ highest freq.

$f = 208, L = 24.5, c = 205, h = 5, n = 726.$

29 → 29
176 → 205
208 ← 413
173

Q2r

a) Conditional Probability: It is the likelihood of an event occurring given that another event has already occurred. Denoted as

$$P(A|B)$$

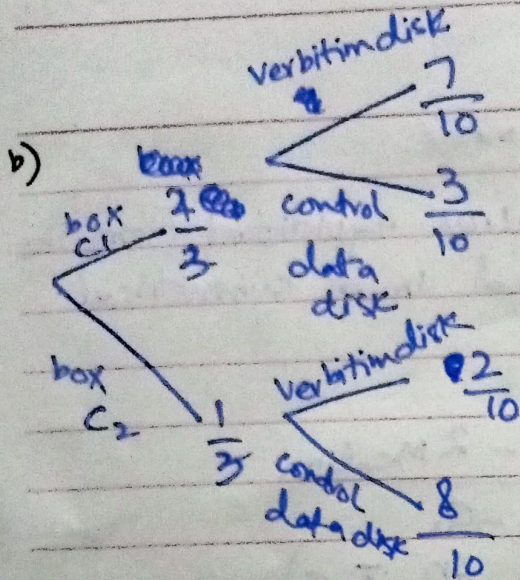
Prob. of A given that B.

formula: $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Simple Events: An event that cannot be broken down into simple components. Consists of a single outcome in sample space, e.g. when rolling a six-sided die, getting 3 is a simple event.

Compound Events: Consists of 2 or more simple events. Involves combination of multiple outcomes, e.g. when rolling a six-sided die, getting an even no (2, 4, 6) is a compound event b/c it combines multiple simple events.

day / date:



1 box selected.
 $P(C_1) = \frac{2}{3}$, $P(C_2) = \frac{1}{3}$

1 disk selected
 let C be disk from control/data disk.

$P(C_1|C) = ?$, $P(C_2|C) = ?$

$$P(C_1|C) = \frac{P(C_1 \text{ and } C) \times P(C_1)}{P(C)} \quad , \quad P(C_2|C) = \frac{P(C_2 \text{ and } C) \times P(C_2)}{P(C)}$$

~~scribbles~~

$$P(C_1|C) = \frac{\left(\frac{2}{3} \times \frac{3}{10}\right)}{\left(\frac{2}{3} \times \frac{3}{10}\right) + \left(\frac{1}{3} \times \frac{8}{10}\right)}$$

$$P(C_1|C) = \frac{2}{7}$$

$$P(C_2|C) = \frac{\left(\frac{1}{3} \times \frac{8}{10}\right)}{\left(\frac{2}{3} \times \frac{3}{10}\right) + \left(\frac{1}{3} \times \frac{8}{10}\right)}$$

$$P(C_2|C) = \frac{4}{7}$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

c)

i. sum is even.

← E ⊕ E ✓

E ⊕ E 0 X

E ⊕ O E X

O ⊕ O 0 ✓

O ⊕ E E X

O ⊕ E E X

O ⊕ O 0 ✓

O ⊕ O 0 X

20 +ve integer let E be even
 let O be odd

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

11, 12, 13, 14, 15, 16, 17, 18, 19, 20.

$$n(E) = 10$$

$$P(E) = \frac{2}{19}$$

$$n(O) = 10$$

$$P(O) = \frac{10}{19}$$

$$P(20 \text{ or } 1E) = \frac{15}{38}$$

$$C_3 = 120$$

$$C_2 = 450$$

$$C_1 = 10$$

$$C_2 = 450$$

$$C_3 = 120$$

$$20 \times C_3 = 1140$$

$$\frac{2}{19} + \frac{15}{38} = \frac{1}{2}$$

KAGHAZ
 www.kaghaaz.pk

1 2 3 4 5 6 7 8 9 10
 11 12 13 14 15 16 17 18 19 20
 day / date:

ii. Product is even.

6 6 6	✓
6 6 0	✓
6 0 6	✓
6 0 0	✓
0 6 6	✓
0 6 0	✓
0 0 6	✓
0 0 0	X

$$P(3 \text{ even}) = \frac{10C_3}{20C_3} +$$

$$P(2 \text{ even}, 1 \text{ odd}) = \frac{10C_2 \times 10C_1}{20C_3} +$$

$$P(1 \text{ even}, 2 \text{ odd}) = \frac{10C_1 \times 10C_2}{20C_3}$$

$$P(\text{Prod is even}) = \frac{17}{19}$$

OR.

$$P(3 \text{ odd}) = \frac{10C_3}{20C_3} = \frac{2}{19}$$

$$P(\text{Prod is even}) = 1 - \frac{2}{19} = \frac{17}{19}$$

Q3:- not done.

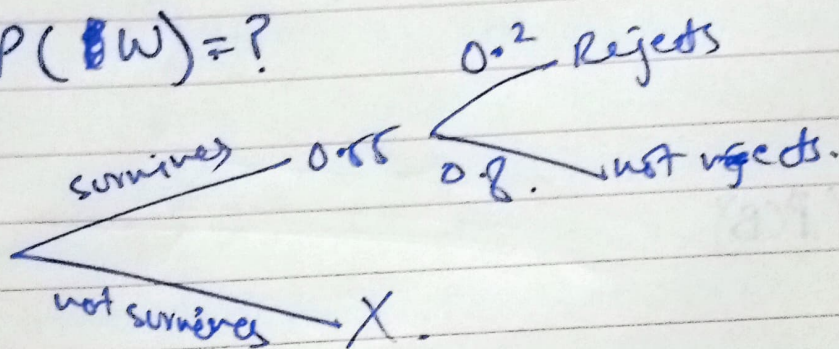
Q4:-

Q4:-

a) $P(S) = 0.55$
~~Rejects~~ (survival)

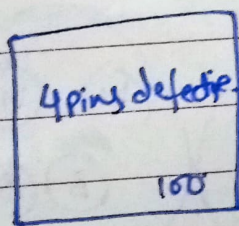
$P(S \cap R) = 0.2$

$P(W) = ?$



$0.55 \times 0.8 = 0.44$

b) $P(D) = 0.05$



$P(X=4) = \frac{100}{C_4} (0.05)^4 (0.95)^{96}$
 $= 0.178$

$P(X=3) = 0.13957$

$P(X=2) = 0.081$

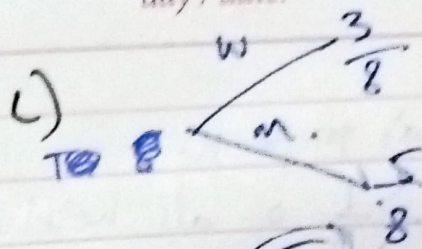
$P(X=1) = 0.031$

$P(X=0) = 5.92 \times 10^{-3}$

$P(X > 4) = 1 - P(X \leq 4)$
 0.43

$1 - 0.43 = 0.57$

day / date:



selection = 5

all men.

$\frac{5C5}{8C5} = \frac{1}{8}$

all women

not possible

Q5 a) google it.

b)

x	1	2	3	4
$f(x)$	$1/10$	$3/10$	A	$1/10$

$$A = ? , \text{Var}(X) = ?$$

$$A = \frac{1}{2}$$

$$E(X) = \sum x f(x)$$

$$= \left(1 \times \frac{1}{10}\right) + \left(2 \times \frac{3}{10}\right) + \left(3 \times \frac{1}{2}\right) + \left(4 \times \frac{1}{10}\right)$$

$$= \frac{13}{5}$$

$$E(X^2) = \sum x^2 f(x)$$

$$= \left(1 \times \frac{1}{10}\right) + \left(4 \times \frac{3}{10}\right) + \left(9 \times \frac{1}{2}\right) + \left(16 \times \frac{1}{10}\right)$$

$$= \frac{37}{5}$$

$$\text{Var} = E(X^2) - (E(X))^2$$

$$= \frac{37}{5} - \left(\frac{13}{5}\right)^2$$

$$= \frac{16}{25}$$

c) not done.